UniDriveVLA: Unifying Understanding, Perception, and Action Planning for Autonomous Driving

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Vision-Language-Action (VLA) models have recently emerged in autonomous driving, with the promise of leveraging rich world knowledge to improve the cognitive capabilities of driving systems. However, adapting such models for driving tasks currently faces a critical dilemma between spatial perception and semantic reasoning. Consequently, existing VLA systems are forced into suboptimal compromises: directly adopting 2D Vision-Language Models yields limited spatial perception, whereas enhancing them with 3D spatial representations often impairs the native reasoning capacity of VLMs. We argue that this dilemma largely stems from the coupled optimization of spatial perception and semantic reasoning within shared model parameters. To overcome this, we propose **UniDriveVLA**, a **Uni**fied **Driv**ing **V**ision-**L**anguage-**A**ction model based on Mixture-of-Transformers that addresses the perception–reasoning conflict via expert decoupling. Specifically, it comprises three experts for driving understanding, scene perception, and action planning, which are coordinated through masked joint attention. In addition, we combine a sparse perception paradigm with a three-stage progressive training strategy to improve spatial perception while maintaining semantic reasoning capability. Extensive experiments show that UniDriveVLA achieves state-of-the-art performance in open-loop evaluation on nuScenes and closed-loop evaluation on Bench2Drive. Moreover, it demonstrates strong performance across a broad range of perception, prediction, and understanding tasks, including 3D detection, online mapping, motion forecasting, and driving-oriented VQA, highlighting its broad applicability as a unified model for autonomous driving.

Date: April 2, 2026
Project: https://xiaomi-research.github.io/unidrivevla
Code: https://github.com/xiaomi-research/unidrivevla

1 Introduction

One expert to understand them all, one expert to perceive them all, one expert to act for all.[1]

Autonomous driving has made significant progress in perception [48, 50, 52], prediction [13, 25], and planning [29, 41] in recent years. Against this backdrop, Vision-Language-Action (VLA) models [22, 33, 47, 98, 119] have emerged as a promising direction, leveraging rich world knowledge and reasoning capabilities to improve driving cognition and planning. However, their effectiveness largely depends on the underlying 2D Vision-Language Models (VLMs) [3, 34, 121]. Because these base models are predominantly pre-trained on internet-scale image-text data, they are not designed for explicit spatial perception, which limits the spatial perception of VLA systems in driving scenarios. To improve spatial perception, recent works [18, 22, 59, 93, 95, 117] have explored enhancing VLA systems with 3D spatial representations, either by aligning structured 2D/3D features with language representations or by directly incorporating spatial tokens into a shared decoder. Yet this direction introduces a fundamental dilemma, as illustrated in Fig. 1: directly adopting 2D VLMs preserves

[1]Inspired by Tolkien's line "One Ring to rule them all, One Ring to find them, One Ring to bring them all."

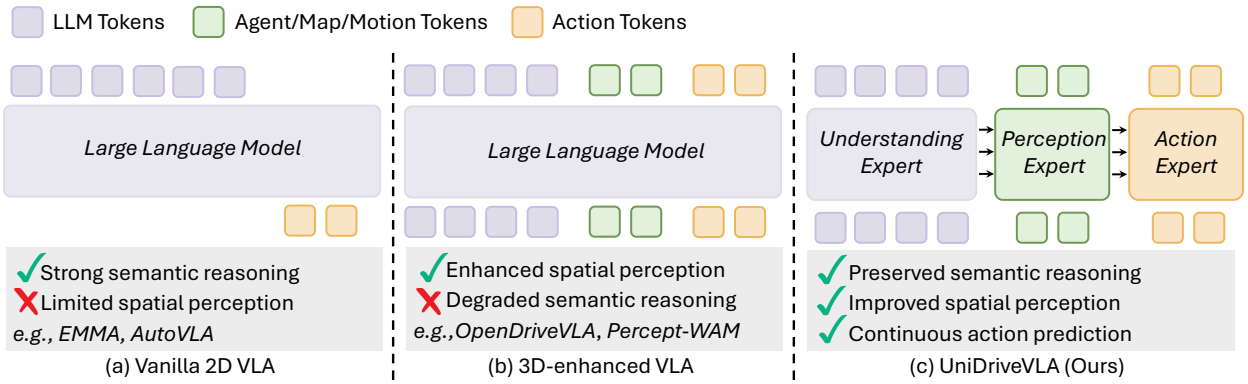

Figure 1 Comparison of VLA paradigms for autonomous driving. (a) Vanilla 2D VLA provides strong semantic reasoning but limited spatial perception. (b) 3D-enhanced VLA improves spatial perception but may degrade semantic reasoning. (c) UniDriveVLA decouples understanding, perception, and action with the Mixture-of-Transformers architecture, achieving both.

native semantic reasoning but yields limited spatial perception, whereas enhancing them with 3D spatial representations improves spatial perception at the expense of native semantic reasoning.

At its core, this dilemma arises not merely from introducing 3D spatial representations, but from jointly optimizing spatial perception and semantic reasoning within shared model parameters. Specifically, although 3D spatial representations can improve spatial perception, jointly optimizing them with semantic reasoning in a shared parameter space introduces representation interference that undermines the native reasoning capacity of VLMs. However, most existing methods integrate spatial and semantic information within unified architectures with shared parameters. Among them, some recent approaches [95, 117] attempt to mitigate this interference by explicitly aligning structured 2D/3D features with language representations, thereby introducing spatial cues while preserving semantic reasoning. Nevertheless, because such alignment data remains far smaller than the internet-scale corpora used for VLM pre-training, these methods can only partially mitigate the conflict and cannot fully eliminate it. Other works [18, 26, 59] instead introduce spatial cues by directly incorporating spatial tokens into a shared decoder, where spatial and semantic tokens are jointly modeled. As we later show in Fig. 2, this observation is further supported by empirical analysis of representation similarity and downstream performance. Consequently, although these methods differ in how spatial information is introduced, they remain unable to fully reconcile spatial perception with semantic reasoning, highlighting the need to decouple these objectives during optimization.

To address this conflict, we propose **UniDriveVLA**, a unified Vision-Language-Action framework that unifies driving understanding, scene perception, and action planning within a single model. The core challenge is that simply decoupling these objectives is insufficient: the model must reduce optimization interference among heterogeneous semantic, spatial, and action tokens, while still preserving the unified decision process and the native semantic reasoning capability of the pre-trained VLM. To this end, UniDriveVLA adopts a Mixture-of-Transformers (MoT) architecture with three specialized experts for understanding, perception, and action, where optimization is decoupled across expert-specific pathways and masked joint attention maintains controlled cross-expert communication for coherent driving behavior. Moreover, instead of introducing dense 3D representations that may disrupt the native reasoning ability of VLMs, we adopt a sparse perception paradigm to extract critical spatial priors directly from 2D VLM tokens, thereby improving spatial perception while preserving semantic reasoning. Finally, to make this decoupled design trainable in practice, we introduce a three-stage progressive training strategy that progressively equips the model with perception and planning capabilities without sacrificing its foundational semantic understanding.

We evaluate **UniDriveVLA** on the widely used nuScenes and Bench2Drive benchmarks. Experimental results show that it achieves state-of-the-art performance in open-loop evaluation on nuScenes and closed-loop evaluation on Bench2Drive. Beyond driving planning, UniDriveVLA also performs well across a broad range of perception, prediction, and understanding tasks, including 3D detection, online mapping, motion forecasting, and driving-oriented visual question answering. These results highlight the advantage of our decoupled design: it improves spatial perception and driving performance while preserving the semantic reasoning ability of

VLMs, demonstrating broad applicability as a unified model for autonomous driving.

- We propose **UniDriveVLA**, a pioneering unified Vision-Language-Action model based on a Mixture-of-Transformers architecture, which mitigates the conflict between spatial perception and semantic reasoning by decoupling their optimization with dedicated experts.
- We introduce a sparse perception paradigm that extracts spatial priors directly from 2D VLM features, together with a three-stage joint training strategy that improves perception and planning while preserving the native reasoning ability of VLMs.
- Extensive experiments on nuScenes and Bench2Drive show that **UniDriveVLA** achieves state-of-the-art performance in open-loop evaluation on nuScenes and closed-loop evaluation on Bench2Drive, while also demonstrating broad applicability across diverse perception, prediction, and understanding tasks.

2 Related Work

2.1 Vision-Language-Action Models for Autonomous Driving.

Recent studies [22, 33, 47, 95, 98, 106, 110, 117] have explored integrating Vision-Language Models (VLMs) into autonomous driving, leveraging their rich world knowledge and reasoning capabilities to better address long-tail scenarios. Dual-system approaches, such as DriveVLM [91] and Senna [42], adopt a fast-slow paradigm that combines VLMs with end-to-end (E2E) driving models [29, 41, 85], where the VLM predicts low-frequency trajectories or provides high-level instructions to guide the E2E model. In contrast to these dual-system frameworks, recent methods have shifted toward unified single-system architectures [16, 22, 33, 46, 47, 76, 102, 113, 117, 119], enabling a single model to perform trajectory planning directly. To satisfy the high-frequency inference requirements of autonomous driving, several studies couple VLMs with action decoders [1, 22, 24, 40, 43, 46, 47, 58, 76, 77, 99, 104, 106] or employ Diffusion Large Language Models (DLLMs) [17, 68, 103, 109, 120] to accelerate reasoning and support continuous action prediction. Recent works [23, 47, 64–66, 75, 79, 98, 103, 105, 114, 119] have also explored reinforcement learning (RL) for fine-tuning VLA models to further improve driving performance and policy alignment.

However, autonomous driving requires robust scene perception to recognize surrounding objects and road structures. Most existing VLA approaches [33, 98, 119] primarily rely on fine-tuning pre-trained open-source models, whose spatial perception remains insufficient for such spatially demanding driving tasks. To improve spatial perception in VLA models, some approaches incorporate BEV encoders [117] or 3D Q-Formers [5, 22, 95] to extract 3D features that are subsequently fed into LLMs. Alternatively, other methods [18, 26, 59] directly inject agent and occupancy tokens into the LLM decoder and jointly optimize the model for both linguistic and spatial perception objectives. Although these strategies enrich the model with 3D spatial information, they may also introduce interference between spatial perception and language reasoning in shared parameters, thereby weakening the effective use of pre-trained world knowledge. In contrast, our work focuses on reconciling spatial perception and semantic reasoning within a unified driving VLA framework.

2.2 Mixture-of-Transformers.

The Mixture-of-Transformers (MoT) [51] paradigm was originally introduced to unify multiple modalities by integrating modality-decoupled experts within a Mixture-of-Experts (MoE) [35] architecture together with decoupled attention mechanisms, thereby improving both modeling capacity and computational efficiency. Subsequent works [19, 63, 73, 80, 86] extended MoT to multimodal understanding and generation tasks, where joint attention enables the incorporation of knowledge from Large Language Models (LLMs) into the generation process while preserving their comprehension capabilities, thereby improving generation performance. Concurrent with the emergence of Mixture-of-Transformers as a general architectural paradigm, π_0 [8] independently adopted a conceptually similar mixture-based transformer design for robotics, integrating vision-language understanding with flow-matching action experts to unify discrete semantic reasoning and continuous action generation. Motivated by MoT's ability to mitigate modality interference through decoupled parameter spaces and computational sparsity, subsequent works following π_0 have adopted similar architectures in Vision-Language-Action (VLA) models for robotics [7, 10, 11, 14, 30, 44, 62, 72, 88, 89, 107] and autonomous driving [6, 31, 46, 98, 106, 123]. However, in the autonomous driving domain, existing approaches [46, 98, 106]

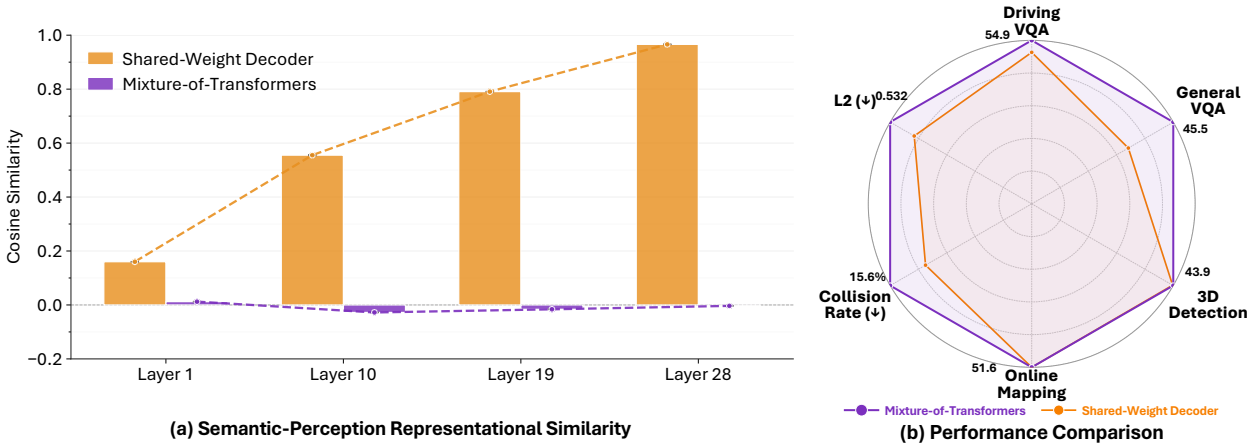

Figure 2 Analysis of representation interference and model performance. (a) Cosine similarity between LLM tokens and perception tokens across layers. In the shared-weight decoder, the similarity progressively increases toward 1, indicating feature collapse into nearly identical representations, whereas MoT maintains low similarity and preserves task decoupling. (b) Performance comparison. By mitigating optimization conflicts, UniDriveVLA consistently outperforms the shared-weight baseline across perception, reasoning, and planning metrics.

largely follow the π_0 design, yet their spatial perception remains limited, which can impair driving performance. In contrast, our work extends the MoT paradigm to autonomous driving by combining expert decoupling with sparse spatial perception, enabling a more effective unification of perception, reasoning, and planning.

2.3 Sparse Perception for Autonomous Driving.

Traditional autonomous driving perception heavily relies on dense 3D representations [71] or BEV grid constructions [45, 48, 50], which introduce computational redundancy. To circumvent this limitation, sparse perception models have emerged as an efficient alternative. These approaches adopt a query-driven paradigm [12], using a set of sparse 3D queries to aggregate multi-view image features. Pioneering works such as DETR3D [97] and PETR [60] first introduced sparse object queries for 3D detection, bypassing dense view transformation. Subsequent advancements, including Sparse4D [53] and SparseBEV [55], further bolstered spatial perception by integrating temporal modeling and adaptive spatio-temporal sampling mechanisms. More recently, this sparse paradigm has been extended to end-to-end autonomous driving systems. Methods such as SparseDrive [85] and SparseAD [122] represent dynamic driving scenes using unified sparse queries and integrate detection, tracking, and planning within a single framework. Nevertheless, existing methods have not yet fully integrated sparse spatial perception with language-driven reasoning in VLA frameworks.

3 Methodology

In this section, we present the methodology of **UniDriveVLA**, with an overview shown in Fig. 3. We begin by formulating the perception–reasoning conflict in monolithic VLA models, where jointly optimizing spatial perception and semantic reasoning within shared parameters leads to representation interference. We then introduce the **UniDriveVLA** architecture based on a Mixture-of-Transformers (MoT) paradigm, which decouples the model into specialized experts for understanding, perception, and planning. Next, we describe a sparse perception mechanism that extracts spatial priors from 2D VLM features and supports downstream planning within the decoupled framework. Finally, we present a three-stage progressive training strategy that enables stable optimization of perception and planning without sacrificing the model's semantic capabilities.

3.1 Problem Definition

The autonomous driving task aims to predict a safe future trajectory T_{traj} from multi-view camera observations $I_{\text{cam}} \in \mathbb{R}^{K\times V\times H\times W\times 3}$, historical trajectory $I_{\text{hist}} \in \mathbb{R}^{T_{\text{hist}}\times 2}$, and navigation command L_{nav}.

$$T_{\text{traj}} = \Phi\big(I_{\text{cam}},\, I_{\text{hist}},\, L_{\text{nav}}\big), \tag{1}$$

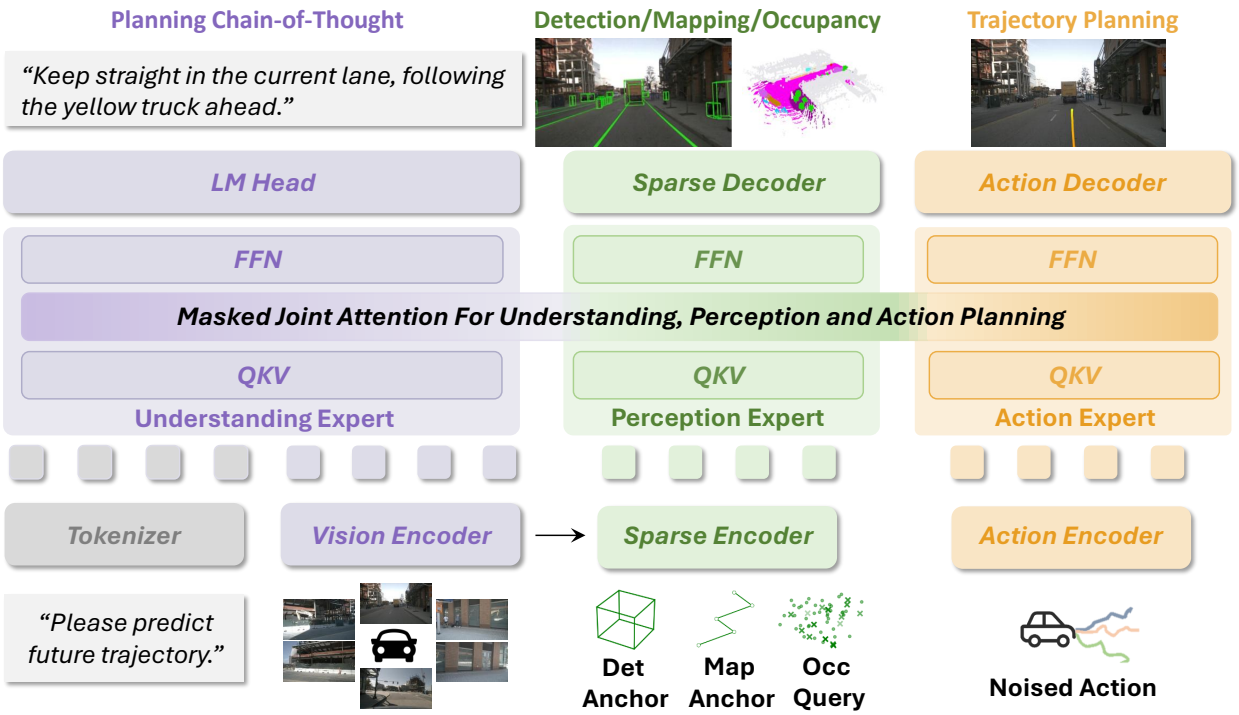

Figure 3 Architecture overview of UniDriveVLA. UniDriveVLA adopts a Mixture-of-Transformers architecture with three specialized experts for driving understanding, scene perception, and action planning. By decoupling heterogeneous tokens into dedicated experts and coordinating them through masked joint attention, the model mitigates optimization conflicts and unifies understanding, perception, and planning within a single framework.

where $T_{\text{traj}} = \{(x_t, y_t)\}_{t=1}^{T}$ denotes the predicted future trajectory.

To improve spatial perception, recent 3D-enhanced VLA models further introduce spatial representations T_{sp}, such as structured 2D/3D features or spatial tokens, into the same shared-weight decoder:

$$T_{\text{traj}} = \Phi\big(I_{\text{cam}},\, I_{\text{hist}},\, L_{\text{nav}},\, T_{\text{sp}}\big). \tag{2}$$

However, although these spatial representations can enhance spatial perception, jointly modeling them with semantic reasoning within shared parameters introduces representation interference between heterogeneous features. As shown in Fig. 2, this interference causes perception and semantic features to become increasingly entangled with depth, which weakens the native reasoning capacity of VLMs and ultimately degrades downstream driving performance.

3.2 UniDriveVLA Architecture

To address the perception–reasoning conflict in monolithic VLA models, **UniDriveVLA** adopts a Mixture-of-Transformers (MoT) [51] architecture that decouples the model into three specialized experts for *driving understanding*, *scene perception*, and *action planning*. Instead of forcing heterogeneous tokens to share a single parameter space, our architecture assigns them to expert-specific pathways while retaining controlled cross-expert interaction through masked joint attention.

Given multi-view images I_{cam}, historical trajectories I_{hist}, and navigation command L_{nav}, UniDriveVLA constructs three groups of tokens. First, a vision-language backbone produces understanding tokens T_{und} from visual observations and driving instructions. Second, a sparse perception module extracts perception tokens T_{per} from visual features to encode spatial priors. Third, an action encoder produces action tokens T_{act} for planning-oriented trajectory modeling, where the action inputs are constructed by standard flow matching interpolation between Gaussian noise and the target velocity sequence.

Within each MoT layer, the three token groups are first processed by expert-specific projections. For each

expert $g \in \{\text{und}, \text{per}, \text{act}\}$, the query, key, and value representations are computed as

$$\mathbf{Q}^g = T_g \mathbf{W}_Q^g,\ \mathbf{K}^g = T_g \mathbf{W}_K^g,\ \mathbf{V}^g = T_g \mathbf{W}_V^g. \tag{3}$$

This expert-specific parameterization decouples understanding, perception, and action into separate parameter subspaces before cross-expert interaction.

To coordinate information flow across experts, we introduce Masked Joint Attention. After expert-specific projection, the representations from all experts are concatenated in the order of understanding, perception, and action:

$$\mathbf{Q} = [\mathbf{Q}^{\text{und}}; \mathbf{Q}^{\text{per}}; \mathbf{Q}^{\text{act}}],\ \mathbf{K} = [\mathbf{K}^{\text{und}}; \mathbf{K}^{\text{per}}; \mathbf{K}^{\text{act}}],\ \mathbf{V} = [\mathbf{V}^{\text{und}}; \mathbf{V}^{\text{per}}; \mathbf{V}^{\text{act}}], \tag{4}$$

and attention is computed globally with a mask matrix $\mathbf{M}$ that controls the visibility pattern:

$$\mathbf{Z} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}. \tag{5}$$

The resulting attention features are then split back into three expert groups,

$$\mathbf{Z} = [\mathbf{Z}^{\text{und}}; \mathbf{Z}^{\text{per}}; \mathbf{Z}^{\text{act}}], \tag{6}$$

where $\mathbf{Z}^{\text{und}}$, $\mathbf{Z}^{\text{per}}$, and $\mathbf{Z}^{\text{act}}$ correspond to the outputs associated with the understanding, perception, and action experts, respectively.

Here, understanding tokens follow causal masking and do not attend to subsequent perception or action tokens, preserving the semantic reasoning capability of the pre-trained vision-language model. Perception tokens attend to preceding understanding tokens to acquire semantic context, while action tokens aggregate both semantic and spatial information for planning. After attention, each expert group is further updated by expert-specific output projection, normalization, and feed-forward transformation:

$$\mathbf{H}^g = T_g + \text{LN}_{\text{attn}}^g(\mathbf{Z}^g \mathbf{W}_O^g),\ \mathbf{O}^g = \mathbf{H}^g + \text{LN}_{\text{ffn}}^g(\text{FFN}^g(\mathbf{H}^g)), \tag{7}$$

where $g \in \{\text{und}, \text{per}, \text{act}\}$. The resulting expert features are then jointly optimized with a unified objective over understanding, perception, and action modeling within a single framework. Specifically, the understanding branch supports autoregressive language modeling, the perception branch is supervised by structured perception tasks, and the action branch is trained with flow-matching-based trajectory generation:

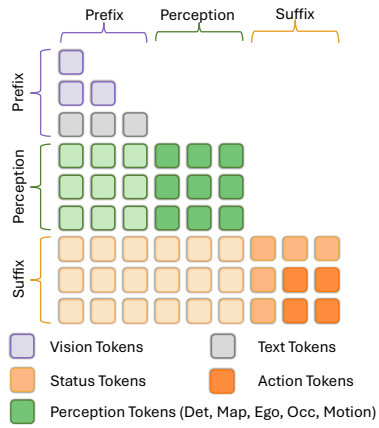

Figure 4 Illustration of Masked Joint Attention.

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{ar}} + \lambda_2 \mathcal{L}_{\text{per}} + \lambda_3 \mathcal{L}_{\text{act}}. \tag{8}$$

In this way, UniDriveVLA unifies understanding, perception, and planning within a single framework while avoiding the optimization conflict of monolithic shared-parameter VLA models.

3.3 Sparse Spatial Perception

Unlike recent approaches [95, 117] that inject dense bird's-eye-view representations into Vision-Language-Action models, UniDriveVLA constructs sparse spatial perception directly from multi-scale 2D visual features. Specifically, projected visual features serve as the geometric evidence for a unified query-based perception module that jointly models detection, online mapping, ego-status estimation, and motion forecasting, rather than assigning each task to an isolated prediction head. Task-specific sparse queries are initialized from instance banks obtained by dataset-level K-Means clustering and are then updated through temporal interaction, intra-task reasoning, inter-task communication, deformable feature aggregation, and task-wise refinement. In this way, the perception branch captures temporal dynamics, task-specific structure, and cross-task dependencies within a unified sparse decoding process. In parallel, occupancy is modeled as an auxiliary latent branch, so

that the perception expert covers heterogeneous yet mutually supportive outputs, including 3D detection, map prediction, ego-status, motion, and occupancy.

To further enrich sparse perception with high-level semantics while preserving the native reasoning behavior of the pre-trained vision-language model, we project the first-pass perception outputs into the hidden space of the perception expert and let them interact with the understanding and action branches through masked joint attention. Concretely, detection, map, ego, motion, and occupancy tokens are lifted into the VLM hidden space, where understanding tokens remain causally masked, while perception and action tokens selectively aggregate preceding semantic and spatial context. As a result, perception tokens are semantically enhanced by the understanding branch and are further aggregated by the action branch to guide planning. The resulting features are then projected back to the sparse perception space and further refined by a subsequent perception decoder, yielding semantically enriched perception outputs that are better aligned with downstream planning. In this way, sparse perception in UniDriveVLA is not merely a one-shot geometric extractor, but a semantically enhanced perception module that collaborates with the understanding and action branches to guide planning, while also providing interpretable perception outputs.

3.4 Three-Stage Progressive Training

To preserve semantic reasoning while progressively acquiring spatial perception and planning capability, we devise a three-stage progressive training strategy for UniDriveVLA, which stabilizes optimization and mitigates catastrophic forgetting.

The first stage aims to anchor the model's semantic reasoning capability through large-scale multimodal pre-training. We follow established paradigms [47] and curate a mixture of driving-specific visual question-answering datasets and general-domain multimodal data. By filtering low-quality driving dialogues and maintaining a dominant sampling ratio for high-quality general data, we preserve the foundational semantic capabilities of the vision-language model.

The second stage aims to incorporate perception and planning supervision into the model through controlled joint optimization. We perform joint optimization over autoregressive language modeling, spatial perception tasks including 3D detection, online mapping, and occupancy prediction, together with flow-matching-based trajectory generation. To preserve semantic reasoning capability during this controlled joint optimization, we apply Low-Rank Adaptation [27] to the vision-language model. Furthermore, we reduce the base learning rate of the VLM parameters by half to suppress aggressive updates that may disrupt semantic reasoning.

The final stage aims to specialize the Perception Expert and Action Expert while preserving the semantic pathway of the vision-language model. We freeze the vision-language model and then fine-tune the Perception Expert and Action Expert while introducing an additional motion prediction objective. This objective is introduced to provide dynamic priors for the action expert and support motion-aware planning.

4 Experiments

4.1 Experimental Setup

Implementation Details. Our framework is built upon Qwen3-VL [3], a vision-language model consisting of a SigLIP-2 [92] vision encoder, an MLP-based vision-language merger, and a Qwen3 LM [4]. All input driving frames are resized to 960×544 to align with the $32\times$ spatial downsampling stride of the Qwen3-VL vision encoder. For the first stage of driving pre-training, following ReCogDrive [47], we fully fine-tune the base VLM for 3 epochs with a learning rate of 4×10^{-5} on a mixture of driving-specific data and general vision-language data. The training mixture is constructed with a driving-to-general data ratio of 3:7, where the general-domain portion mainly comes from FineVision [100]. In the second stage, we jointly train the model for 30 epochs using AdamW [61]. The base learning rate is set to 2×10^{-4}, while the VLM backbone uses a $0.5\times$ learning-rate multiplier, yielding an effective learning rate of 1×10^{-4}. We additionally apply LoRA [27] to the language model and use EMA during training. In the final stage, we freeze the vision-language model and fine-tune the Perception and Action Experts for 15 epochs, while enabling the motion forecasting objective. The base learning rate is set to 1×10^{-4}, and EMA is maintained during training.

Table 1 Open-loop and closed-loop planning results on Bench2Drive. Avg. L2 is computed over predictions within a 2-second horizon at 2 Hz. † Models trained on PDM-Lite, which provides higher-quality expert demonstrations than Think2Drive. Bold numbers indicate the best results among methods trained without PDM-Lite.

Method	Open-loop Metric	Closed-loop Metric			
	Avg. L2 ↓	Driving Score ↑	Success Rate (%) ↑	Efficiency ↑	Comfortness ↑
AutoVLA† [NeurIPS25] [119]	-	78.84	57.73	146.93	39.33
SimLingo† [CVPR25] [76]	-	85.94	66.82	244.18	25.49
R2SE† [T-PAMI26] [57]	-	86.28	69.54	243.89	23.26
AD-MLP [arXiv23] [111]	3.64	18.05	0.00	48.45	22.63
VAD [ICCV23] [41]	0.91	42.35	15.00	157.94	46.01
SparseDrive [ICRA25] [85]	0.87	44.54	16.71	170.21	48.63
GenAD [ECCV24] [116]	-	44.81	15.90	-	-
UniAD [CVPR23] [29]	0.73	45.81	16.36	129.21	43.58
MomAD [CVPR25] [83]	0.82	47.91	18.11	174.91	**51.20**
SeerDrive [NeurIPS25] [112]	0.66	58.32	30.17	-	-
DriveDPO [NeurIPS25] [78]	-	62.02	30.62	166.80	26.79
ThinkTwice [CVPR22] [37]	0.95	62.44	31.23	69.33	16.22
DriveTransformer [ICLR25] [39]	0.62	63.46	35.01	100.64	20.78
DriveAdapter [ICCV23] [36]	1.01	64.22	33.08	70.22	16.01
RAP [ICLR26] [20]	-	66.42	37.27	165.47	23.63
ReCogDrive [ICLR26] [47]	-	71.36	45.45	138.18	17.45
DriveMOE [CVPR26] [106]	**0.38**	74.22	48.64	175.96	15.31
Orion [ICCV25] [22]	0.68	77.74	**54.62**	151.48	17.38
UniDriveVLA	0.72	**78.37**	51.82	**198.86**	11.78

Table 2 Multi-ability results of end-to-end autonomous driving methods on Bench2Drive.

Method	Ability (%) ↑					
	Merging	Overtaking	Emergency Brake	Give Way	Traffic Sign	Mean
AD-MLP [arXiv23] [111]	0.00	0.00	0.00	0.00	4.35	0.87
UniAD [CVPR23] [29]	14.10	17.78	21.67	10.00	14.21	15.55
VAD [ICCV23] [41]	8.11	24.44	18.64	20.00	19.15	18.07
ThinkTwice [CVPR22] [37]	13.72	22.93	52.99	**50.00**	47.78	37.48
DriveAdapter [ICCV23] [36]	14.55	22.61	54.04	**50.00**	50.45	38.33
DriveTransformer [ICLR25] [39]	17.57	35.00	48.36	40.00	52.10	38.60
ReCogDrive [ICLR26] [47]	29.73	20.00	69.09	20.00	71.34	42.03
DriveMOE [CVPR26] [106]	34.67	40.00	65.45	40.00	59.44	47.91
Orion [ICCV25] [22]	25.00	71.11	**78.33**	33.00	**69.15**	**54.72**
UniDriveVLA	**38.75**	**80.00**	50.00	30.00	58.95	51.53

Dataset. We primarily evaluate our framework on two widely adopted autonomous driving benchmarks: nuScenes [9] and Bench2Drive [38], which are used for open-loop and closed-loop evaluation, respectively. The nuScenes dataset consists of 1,000 driving sequences collected in Boston and Singapore, and we use it to evaluate both perception tasks and open-loop planning. Bench2Drive is a large-scale fully annotated benchmark built on the CARLA simulator. It provides 6-view camera inputs at a resolution of 900×1600 and is used for closed-loop driving evaluation. The expert demonstrations in Bench2Drive are officially generated using the Think2Drive [37] paradigm. In addition, we evaluate driving-oriented understanding on DriveBench [101], and assess broader multimodal capability on several general VQA benchmarks, including MMStar [15], MMMU [108], RealWorldQA, AI2D, MME [21], VLMsAreBlind [74], and ChartQA [70].

4.2 Main Results and Ablation Study

Experiments on the Bench2Drive Benchmark. We evaluate UniDriveVLA on Bench2Drive from both planning performance and fine-grained driving ability. As shown in Tab. 1, UniDriveVLA achieves the best Driving Score of 78.37 and the best Efficiency of 198.86 among methods trained without PDM-Lite, while obtaining a competitive Success Rate of 51.82%. In Tab. 2, UniDriveVLA demonstrates strong capability across diverse interactive scenarios, achieving the best performance on Merging with 38.75% and Overtaking with 80.00%, together with a competitive mean score of 51.53%. These results indicate that UniDriveVLA not only improves closed-loop driving quality, but also handles complex interactive behaviors effectively.

Table 3 End-to-end trajectory planning performance on nuScenes. We evaluate planning quality using L2 displacement error and collision rate across two mainstream protocols: ST-P3 [28] and UniAD [29]. * denotes methods using ego-state information. Results for SparseDrive‡ are re-evaluated following the protocol established by GPT-Driver [69].

Method	ST-P3 metrics								UniAD metrics								LLM
	L2 (m) ↓				Collision (%) ↓				L2 (m) ↓				Collision (%) ↓				
	1s	2s	3s	Avg.	1s	2s	3s	Avg.	1s	2s	3s	Avg.	1s	2s	3s	Avg.	
Methods with Ego Status																	
ST-P3* [28]	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71	-	-	-	-	-	-	-	-	-
VAD* [41]	0.17	0.34	0.60	0.37	0.04	0.27	0.67	0.33	-	-	-	-	-	-	-	-	-
UniAD* [29]	-	-	-	-	-	-	-	-	0.20	0.42	**0.75**	0.46	0.02	0.25	0.84	0.37	-
RDA-Driver* [32]	0.17	0.37	0.69	0.40	0.01	**0.05**	0.26	0.10	0.23	0.73	1.54	0.80	**0.00**	0.13	0.83	0.32	LLaVA-7B
BEV-Planner* [49]	0.16	0.32	0.57	0.35	**0.00**	0.29	0.73	0.34	-	-	-	-	-	-	-	-	-
HPP* [56]	0.26	0.37	0.59	0.40	0.02	**0.05**	**0.11**	**0.06**	0.30	0.61	1.15	0.72	0.03	0.07	**0.35**	**0.15**	-
OmniDrive* [95]	**0.14**	0.29	0.55	0.33	**0.00**	0.13	0.78	0.30	-	-	-	-	-	-	-	-	LLaVA-7B
Orion* [22]	0.17	0.31	0.55	0.34	0.05	0.25	0.80	0.37	-	-	-	-	-	-	-	-	LLaVA-7B
FSDrive*† [110]	**0.14**	**0.25**	**0.46**	**0.28**	0.03	0.06	0.21	0.10	**0.18**	**0.39**	0.77	**0.45**	**0.00**	**0.06**	0.42	0.16	Qwen2-VL-3B
AutoVLA* [119]	0.25	0.46	0.73	0.48	0.07	0.07	0.26	0.13	0.33	0.81	1.45	0.86	0.08	0.11	0.85	0.35	Qwen2.5-VL-3B
OpenDriveVLA* [117]	**0.14**	0.30	0.55	0.33	0.02	0.07	0.22	0.10	0.19	0.58	1.24	0.67	0.02	0.18	0.70	0.30	Qwen2.5-VL-3B
UniDriveVLA-Base*	0.23	0.40	0.65	0.43	0.04	0.09	0.18	0.10	0.30	0.69	1.32	0.77	0.03	0.13	0.52	0.23	Qwen3-VL-2B
UniDriveVLA-Large*	0.24	0.40	0.63	0.42	0.03	0.09	0.16	0.10	0.30	0.66	1.25	0.74	0.02	0.18	0.40	0.20	Qwen3-VL-8B
Methods without Ego Status																	
VAD [41]	0.41	0.70	1.05	0.72	**0.03**	0.19	0.43	0.21	-	-	-	-	-	-	-	-	-
UniAD [29]	0.45	0.70	1.04	0.73	0.62	0.58	0.63	0.61	0.59	1.01	1.48	1.03	0.16	0.51	1.64	0.77	-
ELM [118]	-	-	-	-	-	-	-	-	**0.34**	1.23	2.57	1.38	0.12	0.50	2.36	0.99	BLIP2-2.7B
OccWorld [115]	0.39	0.73	1.18	0.77	0.11	0.19	0.67	0.32	0.52	1.27	2.41	1.40	0.12	0.40	2.08	0.87	GPT3-like
BEV-Planner [49]	0.30	0.52	0.83	0.55	0.10	0.37	1.30	0.59	-	-	-	-	-	-	-	-	-
HPP [56]	0.41	0.61	0.86	0.63	**0.03**	0.08	0.24	0.12	0.48	0.91	1.54	0.97	0.03	0.17	0.68	0.29	-
SparseDrive‡ [85]	0.28	0.53	0.84	0.55	**0.03**	**0.05**	**0.15**	**0.08**	0.38	0.92	1.66	0.99	**0.02**	**0.08**	**0.53**	**0.21**	-
OmniDrive [95]	0.40	0.80	1.32	0.84	0.04	0.46	2.32	0.94	-	-	-	-	-	-	-	-	LLaVA-7B
FSDrive [110]	0.28	0.52	0.80	0.53	0.06	0.13	0.32	0.17	0.40	0.89	1.60	0.96	0.07	0.12	1.02	0.40	Qwen2-VL-3B
UniDriveVLA-Base	0.28	0.51	0.82	0.54	0.08	0.13	0.31	0.17	0.37	0.89	1.62	0.96	0.08	0.27	0.88	0.41	Qwen3-VL-2B
UniDriveVLA-Large	**0.27**	**0.49**	**0.77**	**0.51**	**0.03**	0.10	0.21	0.11	0.36	**0.83**	**1.50**	**0.90**	**0.02**	0.23	0.55	0.27	Qwen3-VL-8B

Table 4 Comparison of perception, mapping, and motion prediction performance on the nuScenes validation dataset.

Method	Detection		Map				Motion	
	mAP↑	NDS↑	AP_{ped} ↑	$AP_{divider}$ ↑	$AP_{boundary}$ ↑	mAP↑	minADE(m) ↓	minFDE(m) ↓
UniAD [CVPR23] [29]	0.380	0.359	–	–	–	–	0.710	1.020
VAD [ICCV23] [41]	0.276	0.397	0.406	0.515	0.506	0.476	–	–
SparseDrive [ICRA25] [85]	0.418	0.525	0.499	**0.570**	**0.584**	0.551	**0.600**	**0.960**
EgoFSD [ICRA26] [84]	0.410	0.528	**0.549**	0.557	0.573	0.560	–	–
HiP-AD [ICCV25] [87]	**0.424**	**0.535**	–	–	–	**0.571**	0.610	–
UniDriveVLA-Base	0.397	0.434	0.462	0.556	0.543	0.520	1.396	2.289
UniDriveVLA-Large	0.407	0.460	0.491	0.556	0.557	0.535	1.264	2.121

Planning Results on the nuScenes Benchmark. Tab. 3 presents a comprehensive comparison of end-to-end trajectory planning performance on nuScenes under two evaluation protocols, namely ST-P3 and UniAD. Overall, UniDriveVLA achieves competitive results in both the with-ego and without-ego settings. In particular, under the more challenging setting without ego-state inputs, UniDriveVLA-Large attains the best trajectory accuracy, achieving the lowest average L2 error under both the ST-P3 and UniAD protocols. In the with-ego setting, UniDriveVLA also remains competitive against recent VLA methods such as AutoVLA and FSDrive, as well as strong end-to-end driving baselines including UniAD and VAD. These results suggest that integrating sparse perception priors into a vision-language-action framework can substantially improve open-loop trajectory prediction, especially when explicit ego-state information is unavailable.

Perception Results on the nuScenes Benchmark. Beyond planning, we evaluate the intermediate perception capabilities of UniDriveVLA on the nuScenes validation set. As shown in Tab. 4, UniDriveVLA achieves competitive detection and mapping performance. In particular, UniDriveVLA-Large reaches a detection mAP of 0.407 and an NDS of 0.460, while achieving a map mAP of 0.535. Compared with earlier end-to-end driving methods such as VAD, our model shows clear improvements in both detection and mapping quality, indicating that the proposed sparse perception design can provide a solid spatial foundation for downstream planning. Although its motion prediction accuracy remains behind specialized baselines, these results show

Table 5 Ablation on nuScenes planning. L2 and CR denote average L2 error and collision rate, respectively.

Components						Metrics	
Baseline	Ego	Det	Map	Occ	Motion	L2 ↓	CR (%) ↓
✓						0.75	0.27
✓	✓					0.61	0.21
✓	✓	✓				0.58	**0.10**
✓	✓	✓	✓			0.58	0.14
✓	✓	✓	✓	✓		**0.53**	0.14
✓	✓	✓	✓	✓	✓	0.54	0.17

Table 6 DriveBench comparison. † ReCogDrive is pretrained only, without action training.

Method	Percep.	Predict.	Plan.	Behav.	Avg.
LLaVA-1.5 [54]	23.22	22.02	29.15	13.60	22.00
InternVL2 [90]	32.36	45.52	53.27	54.58	46.43
Qwen2-VL [94]	30.13	49.35	61.30	51.26	48.01
DriveLM [81]	16.85	44.33	68.71	42.78	43.17
Dolphins [67]	9.59	32.66	52.91	8.81	25.99
GPT-4o [34]	35.37	**51.30**	**75.75**	45.40	51.96
ReCogDrive† [47]	**64.95**	49.34	70.20	42.36	**56.71**
UniDriveVLA	36.78	43.13	66.98	**60.97**	51.97

Table 7 Performance comparison between a shared-weight decoder and Mixture-of-Transformers (MoT) across understanding, perception, and planning tasks. General VQA is averaged over general multimodal benchmarks, DriveBench is used for driving-oriented understanding, and perception and planning metrics are reported on nuScenes.

Architecture	Understanding		Perception		Planning	
	General VQA (%) ↑	DriveBench (%) ↑	Det (NDS) ↑	Map (mAP) ↑	L2 (m) ↓	CR (%) ↓
Shared-Weight Decoder	31.1	50.8	0.437	**0.516**	0.641	0.175
Mixture-of-Transformers	**45.5**	**54.9**	**0.439**	**0.516**	**0.533**	**0.140**

Table 8 Comparison of UniDriveVLA with general-purpose and driving-specific models on general multimodal understanding benchmarks.

Model	Params	MMStar↑	MMMU↑	RealWorldQA↑	AI2D↑	MME↑	VLMsAreBlind↑	ChartQA↑
Qwen2.5-VL [2]	7B	63.7	51.1	**69.0**	82.5	2317	40.3	82.5
Qwen3-VL [4]	8B	63.0	52.8	**69.0**	83.2	2364	**61.9**	83.8
InternVL3 [121]	8B	**69.1**	54.8	67.8	**83.9**	**2393**	42.0	86.6
InternVL3.5 [96]	8B	66.3	57.3	63.0	82.3	2357	45.4	**87.1**
GPT-5 Nano [82]	—	41.3	**57.6**	60.7	65.7	-	40.2	48.6
UniDriveVLA	8B	43.3	47.3	49.9	76.3	1876	26.6	76.3

that UniDriveVLA maintains meaningful multi-task perception capability within a unified VLA framework.

Ablation on Planning Components. We progressively incorporate ego-state, detection, mapping, occupancy, and motion components into the planning framework, and report the results in Tab. 5. Starting from the baseline planner, introducing ego-state yields a substantial improvement in both trajectory accuracy and collision rate. Adding detection further reduces the collision rate to 0.10, indicating that explicit object-level perception is particularly beneficial for safe planning. Incorporating occupancy brings the best L2 error of 0.53, suggesting that dense spatial context is especially helpful for trajectory prediction. In contrast, adding mapping and motion does not lead to further gains in the current setting. We attribute this mainly to the limited headroom of open-loop planning on nuScenes, where improvements are already marginal and harder to realize.

Driving Scene Understanding. We evaluate UniDriveVLA on DriveBench to assess driving-oriented scene understanding, and further report results on several general VQA benchmarks to examine whether the model preserves multimodal capability after driving adaptation. As shown in Tab. 6, UniDriveVLA achieves strong performance on DriveBench, demonstrating driving-oriented understanding and behavior reasoning.

Effect of MoT Decoupling. Tab. 7 compares a shared-weight decoder with the proposed Mixture-of-Transformers. MoT consistently improves understanding, perception, and planning performance, with especially clear gains on trajectory prediction and collision reduction. These results support our claim that expert decoupling effectively mitigates the perception–reasoning conflict in unified driving VLA models.

General Visual Capabilities. A key challenge in driving-oriented adaptation is preserving the general multimodal capability of the foundation model. As shown in Tab. 8, although UniDriveVLA does not match

strong general-purpose VLMs on these benchmarks, it retains meaningful performance across a diverse set of general visual understanding tasks, achieving 49.9 on RealWorldQA, 76.3 on AI2D, and 76.3 on ChartQA. These results suggest that our driving adaptation does not collapse the model's multimodal reasoning ability, and that the adopted training strategy preserves a level of visual capability beyond autonomous driving.

5 Conclusion

In this work, we present **UniDriveVLA**, a unified Vision-Language-Action framework for autonomous driving that unifies understanding, perception, and planning within a single model. We identify a fundamental perception–reasoning conflict in existing monolithic VLA models, where jointly optimizing spatial perception and semantic reasoning within shared parameters leads to representation interference. To address this issue, we introduce a Mixture-of-Transformers architecture with dedicated Understanding, Perception, and Action experts, enabling controlled cross-expert interaction while reducing optimization conflict. We further adopt a sparse perception mechanism that extracts spatial priors directly from multi-scale 2D visual features, together with a three-stage progressive training strategy that improves perception and planning while preserving semantic reasoning capability. Extensive experiments on perception, understanding, and planning benchmarks demonstrate the effectiveness of UniDriveVLA as a unified model for autonomous driving. Moreover, the proposed decoupled VLA design may be further extended to robotic manipulation scenarios that require both structured spatial perception and high-level semantic reasoning.

Acknowledgments

We use publicly available datasets, models, and code resources, including DriveLM, LingoQA, nuScenes, Bench2Drive, and other public research resources. We confirm that all such resources are used solely for academic research purposes and not for any commercial activity, in accordance with their respective licenses and terms of use.

We also thank Tianheng Cheng, Haoyu Fu, Lunbin Zeng, Hongyuan Tao, and Haochen Liu for their valuable discussions and helpful exchanges.

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